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PALEOMAGNETIC EVIDENCE FOR A PLEISTOCENE COUNTERCLOCKWISE ROTATION OF THE SANT'ARCANGELO BASIN, SOUTHERN ITALY

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Abstract. New paleomagnetic results from 182 cores sampled at 15 sites in Pliocene and Pleistocene clayey units in the Sant'Arcangelo basin (southern Italy) and its surroundings are presented. They indicate that this area underwent a counterclockwise rotation of 22° after the deposition of the Lower Pleistocene clays.

Introduction

During the past two decades many paleomagnetic studies have been performed in various structural units in the peri-Tyrrhenian orogenic belt. Most of these studies were carried out in Mesozoic units, for which various amounts of counterclockwise rotation in the southern Apennines and clockwise rotation in the Sicilian nappes were found [Manzoni, 1975; Schult, 1976; Channell et al., 1980; Incoronato et al., 1985; Incoronato and Nardi, 1989]. However, since the deformation that produced the present-day structural setting of the peri-Tyrrhenian belt was mostly post-Tortonian, it should be useful to investigate the paleomagnetism of Neogene geological units contemporaneous with the main geodynamic events. As a matter of fact, some results from southern Italy [Tauxe et al., 1983; Aifa et al., 1988; Scheepers et al., 1991] show that vertical axis rotations occur even in Pleistocene rocks. In order to determine vertical axis rotations characterizing the most recent phases of geodynamic evolution in the southern Apennines, a paleomagnetic analysis of Plio-Pleistocene clayey units outcropping in an area with widespread evidence of Pleistocene tectonics, the Sant'Arcangelo basin, was carried out.

Geological setting of the Sant'Arcangelo basin

The Sant'Arcangelo basin is located in the southern Apennines, on the inner front of the allochthonous thrust sheets (Figure 1). It is structurally defined as a "piggy-back" basin [Hippolyte et al., 1991] and is filled with thick Plio-Pleistocene terrigenous sequences referred to distinct sedimentary phases [Vezzani, 1967; Caldara et al., 1988]. An Upper Pliocene-Lower Pleistocene sequence widely prevails in outcrop. It is represented by a basal marly clayey unit (dated at the NN19 nannofossil zone [Hippolyte et al., 1991]), 600-700 m thick, passing westward and upward to sands, and finally to conglomerates [Vezzani, 1967]. This sequence is weakly deformed and defines a broad NW-SE syncline (with bedding dip always < 30°) with some minor folds. An older Upper Pliocene sequence (with a clayey unit dated at the Globorotalia puncticulata zone [Vezzani, 1967] and at the NN14 to NN16 nannofossil zone [Hippolyte et al., 1991]) outcrops

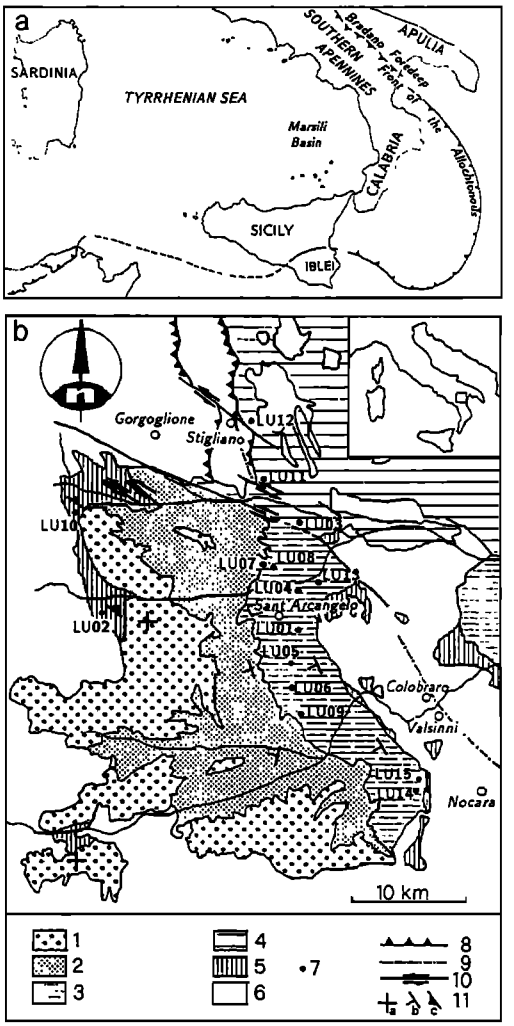


Fig. 1: a) Geographical sketch map of Tyrrhenian Sea surroundings; b) Geological sketch map of the Sant'Arcangelo basin (from Hippolyte et al. [1991] simplified and modified). Location of the sampled area is shown in the inset to the upper right. Upper Pliocene-Lower Pleistocene cycle: (1) Conglomerates - (2) Sands - (3) Marly clays; (4) Pliocene and Pleistocene; (5) Pliocene; (6) Undifferentiated allochthon; (7) Sampling sites; (8) Overthrust front; (9) Anticline axis; (10) Strike-slip fault; (11) Bedding attitude; a: dip <10°, b: dip 10° - 30°, c: dip > 30°.

only at the basin's western and eastern edges and shows a strong tilting (up to 65°) and a rather complex pattern of deformation. The basin is bounded to the north by a WNW-ESE left-lateral strike-slip fault. North of this fault eastward verging thrust sheets overlie Upper Pliocene [Lentini, 1979] clays. Previous studies [Cal-

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dara et al., 1988; Hippolyte et al., 1991] yielded evidence for synsedimentary tectonics; mesostructural analysis detected a single Pleistocene compressive episode with a maximum principal stress trending N70° (both inside and outside the basin). The occurrence of Lower Pleistocene fan-delta or littoral sediments in the basin [Caldara et al., 1988] presently up to 900 m above sea level indicates a high rate of Pleistocene vertical uplift.

Strike-slip fault systems in the southern Apennines were recently evidenced, defining crustal blocks that may have rotated around vertical axes. Turco and Malito [1988] proposed a kinematic model in which they explicitly indicated wrench-fault tectonics as a cause for the origin and possible rotation of the Sant'Arcangelo basin.

Paleomagnetic sampling and NRM measurements

Sampling was done with an electrical drill. The corer was a 25-mm barrel with a sintered diamond cutting head cooled by water. Only the Plio-Pleistocene clayey units were sampled: 182 oriented cores were collected from 15 sampling sites (Figure 1). Two sites are from the Upper Pliocene clays of the basin's western edge, ten are from the Upper Pliocene-Lower Pleistocene clays of the eastern limb of the main syncline and three are from Upper Pliocene clays north of the basin. Two of these latter sites are very close to the front of the easternmost thrust of the Stigliano area.

All measurements were made at the new paleomagnetic laboratory of the Istituto Nazionale di Geofisica, located in L'Aquila, Italy. From each core 1 to 4 standard specimens (22 mm high) were sliced. Natural remanent magnetization (NRM) was measured using a JR-4 spinner magnetometer. Stepwise thermal demagnetization was carried out on one specimen from each core, for which the low-field magnetic susceptibility was measured at each heating step, using a KLY-2 Kappabridge.

A pilot specimen from each site was demagnetized in order to identify the temperature at which main magnetic changes could take place. Such pilot specimens were heated up to 400°C in 10 steps; significant increases in the magnetic susceptibility, which could indicate the formation of new mineralogic phases, generally occurred between 310°C and 340°C and were associated with drastic changes in the magnitude and direction of the residual magnetization. On the basis of the behaviour of the pilot specimens the remaining specimens were thermally demagnetized in 7 - 9 steps from room temperature up to 340°C - 360°C, at which point their magnetic properties had already undergone a major change. The initial NRM intensities were generally of order of 10^{-3}A/m, ranging from 1.1 x 10^{-4}A/m to 1.3 x 10^{-1}A/m. During thermal demagnetization the residual NRM reached half of the initial intensity at about 250°C and at the temperatures corresponding to the major magnetic changes it was on the order of 10% of the initial intensity for most of the specimens. The anisotropy of magnetic susceptibility (AMS) of the collected samples was also analyzed but the results will not be shown here in detail.

Data analysis

All sites but one include at least one specimen with a single stable component of magnetization isolated during stepwise thermal demagnetization. These specimens, usually those with the higher NRM intensities, give a straight line toward the origin in a Zijderveld diagram, after the removal of a soft viscous component (at about 150°C - 200°C), if present (Figure 2a). Each site includes also specimens with two components of magnetization with unblocking spectra mostly overlapping. Their demagnetization path results in a "remagnetization circle" on a stereonet (Figure 2b). Best-fit lines and planes were estimated using the technique of principal component analysis described by Kirschvink [1980]. Steps containing a viscous component or above the major change in magnetic properties were discarded from the analysis. The mean direction for each site was evaluated using the combined analysis of stable endpoints and remagnetization circles according to the method of McFadden and McElhinny [1988] (Figure 2c). Paleomagnetic data were

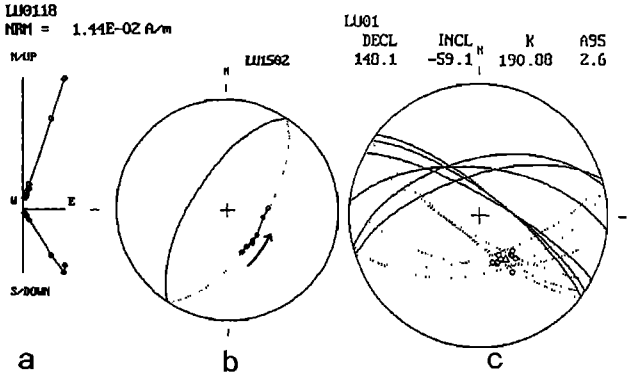


Fig. 2: a) Representative vector diagram of a specimen (LU0118) with a single stable component of magnetization. Full circles - horizontal projection; open circles - vertical projection. b) Representative demagnetization path for a specimen (LU1502) which defines a remagnetization circle. Solid line - lower hemisphere; open circles and dashed line - upper hemisphere. c) Example of convergence for remagnetization circles and stable endpoints for a site (LU01). Open triangle - convergence point.

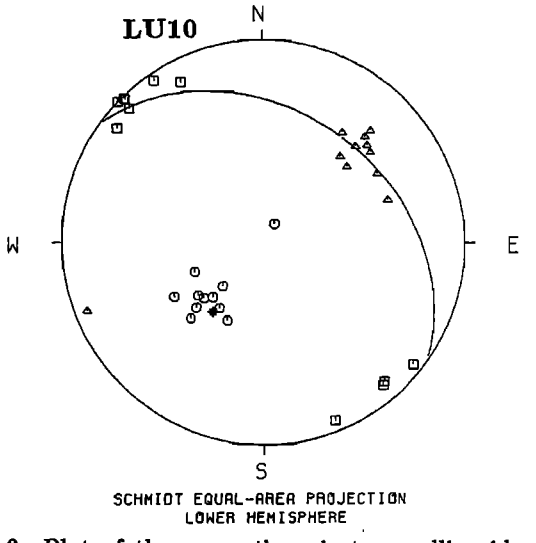


Fig. 3: Plot of the magnetic anisotropy ellipsoid axes obtained for the specimens from site LU10.
□ - K_{max}; △ - K_{int}; ○ - K_{min}. The bedding plane and its pole (*) are also drawn.

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considered both before and after correction for structural tilt of the strata. Since bedding was not accurately measurable at each sampling site, its attitude was determined "a posteriori" by AMS laboratory results. AMS analysis showed that the magnetic fabric of the sampled units is typical of sediments at the very first stage of deformation (or even undeformed). Therefore it is possible to assume K_{min} to be normal to bedding. Comparison of the bedding attitude determined by AMS results with those directly measured in the field (for the sites where it was evident; see i.e. Figure 3), or reported on published geological maps [Vezzani, 1967; Lentini, 1979] gives confidence to this assumption. For site LU02 a within site variable bedding attitude was recognized; a different tilt correction was used for each core. The mean paleomagnetic directions obtained from each site, before and after bedding correction, are listed in Table 1 and plotted on an equal-area projection in Figure 4. Sites LU11 and LU12 were discarded from the overall mean, calculated using Fisher [1953] statistics, because they are located at the thrust front (north of the basin). This structural position is thought to cause the observed deviation of their site mean directions from the main cluster. All sites but LU02 show reversed paleomagnetic directions. Site LU02, after bedding correction, gives a normal paleomagnetic direction antipodal to the cluster of the reversed sites. Correction for structural tilting results in a better grouping of the data. Fold tests, both by the McElhinny [1964] and the McFadden and Jones [1981] methods, are positive at the 95% level of confidence (Table 2).

Table 1: Mean paleomagnetic directions calculated at each site from combined analysis of remagnetization circles and stable endpoints

Sites	$N(n,c)$	D_1/I_1	K_1	$\alpha95_1$	D_2/I_2	K_2	$\alpha95_2$
LU01	17(11,6)	163.2/-57.5	190.9	2.6	148.1/-59.1	190.9	2.6
LU02	14(11,3)	242.0/55.3	11.1	12.5	344.1/51.1	26.9	7.9
LU03	10(0,10)	163.3/-50.5	30.1	9.9	158.0/-56.3	30.1	9.9
LU04	9(5,4)	165.9/-56.5	304.5	3.1	148.7/-59.1	304.5	3.1
LU05	10(5,5)	188.0/-56.6	39.4	8.0	156.5/-61.9	39.4	8.0
LU06	9(6,3)	188.5/-52.6	40.6	8.4	169.8/-57.2	40.6	8.4
LU07	11(4,7)	192.3/-75.4	210.7	3.3	165.5/-63.7	210.7	3.3
LU08	11(3,8)	174.1/-81.8	157.9	3.8	144.7/-62.3	157.9	3.8
LU09	10(6,4)	183.2/-37.3	27.0	9.7	169.5/-49.7	27.0	9.7
LU10	8(5,3)	109.3/-53.4	93.7	5.9	157.4/-49.6	93.7	5.9
LU11*	10(1,9)	136.7/-46.6	18.0	12.6	195.2/-60.9	18.0	12.6
LU12*	8(2,6)	142.8/-24.8	28.7	11.4	133.5/-27.2	28.7	11.4
LU13	11(7,4)	144.7/-63.2	117.3	4.3	144.4/-56.2	117.3	4.3
LU14	9(2,7)	162.0/-29.0	18.3	13.2	149.4/-47.8	18.3	13.2
LU15	6(1,5)	186.6/-36.1	27.4	15.4	170.8/-51.0	27.4	15.4
Mean:	13	168.3/-56.1	17.8	9.2	157.8/-56.1	111.2	3.7

*: Sites rejected for their structural position (see text). The overall mean is calculated using Fisher [1953] statistics; for site LU02 the antipodal (reversed) direction was considered. N:number of samples (n=number of direct observations; c=number of remagnetization circles); D, I: declination and inclination; K: precision parameter; α_{95}: half-angle of cone of 95% confidence about the mean direction; 1: before bedding correction; 2: after bedding correction.

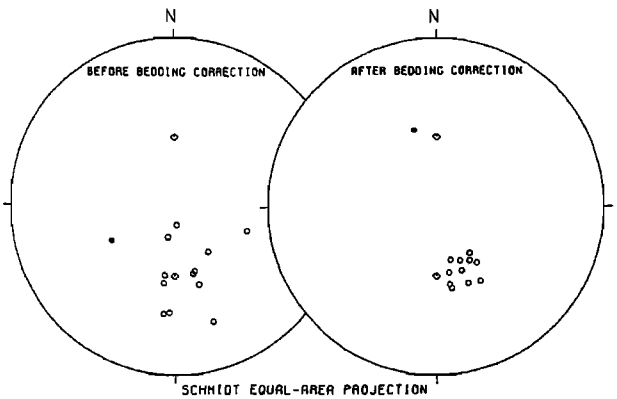


Fig. 4: Equal-area projection of the sites mean directions listed in Table 1. Full circles - normal magnetization; open circles - reversed magnetization; ◇ - present and reversed (antipodal) directions of the geomagnetic field.

The results shown above can be summarized as follows: 1) Positive fold tests and antipodal normal (LU02) and reversed (all other sites) site mean directions suggest that the obtained paleomagnetic vectors are a statistic estimate of a primary component of magnetization. That is, the overall mean direction reflects the local geomagnetic vector at the time of deposition (for Upper Pliocene-Lower Pleistocene time it ought to be around D = 180°, I = −56° for a reversed geomagnetic field). 2) The mean direction obtained from the statistical analysis of 13 sites is D = 157.8°, I = −56.1° (α_{95} = 3.7°), after bedding correction, indicating a counterclockwise rotation of about 22° of the Sant'Arcangelo basin. Flattening of the paleomagnetic vectors is absent or negligible. 3) No significant difference has been detected for Pliocene and Pleistocene sites, suggesting that the observed counterclockwise rotation occurred after the deposition of the Lower Pleistocene clays. Moreover, the consistency, inside and outside the basin, of the paleo-stress directions reconstructed by meso-structural analysis [Hippolyte et al., 1991] seems to exclude a differential rotation of the basin with respect to its surroundings. In fact, site LU03 shows the same rotation as the sites located within the basin and a similar rotation is possible at one (LU12) of the two sites at the thrust front.

Table 2: Fold tests: a) following McElhinny [1964] (all sites except LU11 and LU12); b) following McFadden and Jones [1981] (2 sites with opposite bedding dip)

a)

N	K_1	K_2	K_2/K_1	$F_{(24,24)}$
13	17.8	111.2	6.25	1.98

b)

K_a/K_b	$F_{(14,16)}$	R_a	R_b	R_{t1}	FT_1	R_{t2}	FT_2	LV
2.31	2.44	7.941	8.840	15.859	4.094	16.733	0.218	0.221

N: number of sites; K : precision parameter (see Table 1); $F_{(x,y)}$: limiting value derived from F tables with x and y degrees of freedom at the 5% significance level; a : site LU10 (dipping SW); b : site LU06 (dipping NE); t : sites LU10 and LU06 combined; 1 : before bedding correction; 2 : after bedding correction; R : length of the resultant vector; FT : Fold Test value = $[R_a+R_b-R^2/(R_a+R_b)]/2(N'-R_a-R_b)$; LV : Limiting Value at the 5% significance level = $(1/p)^{1/(N'-2)}$ - 1; N' = 17.

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Conclusions

Paleomagnetic data from Plio-Pleistocene clays of the Sant'Arcangelo basin yield evidence for a 22° Pleistocene counterclockwise rotation. This rotation is almost homogeneous in the whole basin and probably occurs also outside its northern edge.

The data presented here confirm the rotation postulated in the model of Turco and Malito [1988] for the Sant'Arcangelo basin but additional paleomagnetic data are necessary to provide a quantitative test for the model. However, paleomagnetic results recently reported from other areas in southern Italy describe a pattern of Pleistocene rotation that seems to be of regional significance. These data indicate that: 1) the Matera area (Bradano foredeep) experienced a counterclockwise rotation quite similar to the one here found for the Sant'Arcangelo basin [Scheepers et al., 1991]; 2) a uniform clockwise rotation of 15° is widely recognized both in Calabria and in Sicily (excluding the Iblei foreland) [Tauxe et al., 1983; Aifa et al., 1988; Scheepers et al., 1991]. In my opinion southeastward migration of rifting and spreading processes in the Tyrrhenian Sea (the Marsili basin has opened during the past 2 million years at a full spreading rate of 3.5 cm/yr, subsiding at a rate of 700 m/My [Kastens et al., 1988]) may be associated with the observed pattern of Pleistocene rotations (counterclockwise in the southern Apennines and clockwise in Calabria and Sicily), resulting in an increase in the peri-Tyrrhenian belt curvature.

Finally, it should be stressed that the amount of rotation found for the Pleistocene of the Sant'Arcangelo basin is of the same order of magnitude as the total amount of rotation detected since the Mesozoic for some different structural domains in the Italian peninsula (see Lowrie [1986] for a review). Since it is well known that the present complex structure of the Apennines is due to diachronous geodynamic events (rotation of the Sardinia-Corsica microplate, progressive eastward migration of the thrust belt-foredeep-foreland system and opening of the Tyrrhenian Sea) it is likely that the rotations detected in the Mesozoic formations are the result of competing phenomena which can only be investigated with detailed paleomagnetic analysis, from the recent to the old, with good stratigraphic control.

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